

The Constants of Nature

$\epsilon_h:\text{K}=\frac{\text{kelvin}}{\text{K}}=\frac{\left(\frac{T_p}{m_p}\right)}{\text{K}}\left(1-14\,C_U^{-4}\,\mathcal{J}_\theta\right)\boxtimes$	$m_{de}=\frac{3s}{\mathcal{K}_3\,\mathcal{K}_4}\left(\frac{16}{4\pi}\left(m_n+m_+\right)\left(1+\frac{Im(\omega_1)}{\mathcal{K}_2}\right)\left(\frac{L_2}{14}\right)\mathcal{K}_r^{-3}\boxtimes\right.$	$\mathcal{E}_p=\frac{\text{joule}}{\text{eV}}\left(\frac{L_p^2\,m_e}{t_p^2}\right)\left(1-\frac{1}{2}Re\left(i^{l^{''}}\right)\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$\frac{\mu_N}{hc}=\frac{\mathcal{K}_1}{4\pi}\left(\frac{t_p}{l_p}\frac{q_p}{m_e}\right)\left(1+\sqrt{\frac{4}{D_{00}}}\mathcal{K}_1\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$c=\left(\frac{L_2}{T_p}\right)\left(1+\sqrt{G_{G1}}\left(\mathcal{K}_1-\mathcal{K}_2\right)\boxtimes\right)$	$F_{90}=\text{farad}\left(1+\frac{C_d}{6}\left(\mathcal{K}_1+\mathcal{K}_3+\mathcal{K}_4\right)\boxtimes\right)$	$K_j=\frac{\mathcal{K}_1}{\pi}\left(\frac{t_p\,q_p}{l_p^2\,m_p}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathcal{K}_1^2+\mathcal{K}_2^2\right)\boxtimes\right)$	$g_{de}=\frac{2\pi}{8K}\left(1+\frac{16}{4\pi}\left(\mathcal{K}_1^2+\mathcal{K}_3^2+\mathcal{K}_4^2\right)\boxtimes\right)$
$\text{K}:\epsilon_h=\frac{\text{kelvin}}{\mathcal{K}_1^4}=\frac{\left(\frac{t_p}{l_p}\frac{m_e}{m_e}\right)}{\mathcal{K}_1^4}\left(1+14\,C_U^{-4}\,\mathcal{K}_\theta\right)\boxtimes$	$m_+=\frac{1}{\mathcal{K}_3\,\mathcal{K}_4}\sqrt{\frac{16}{4\pi}}\left(m_n+m_+\right)\left(1+\left(\frac{16}{2\pi}\right)\mathcal{K}_r^{-3}\boxtimes\right)$	$R_K=\frac{2\pi}{\mathcal{K}_1^2}\left(\frac{L_p^2\,m_p}{t_p\,q_p^2}\right)\left(1+\frac{1}{2}Re\left(i^{l^{''}}\right)\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$\frac{\mu_N}{hc}=\frac{\mathcal{K}_1}{4\pi}\left(\frac{t_p\,q_p}{l_p}\frac{q_p}{m_e}\right)\left(1+\sqrt{\frac{4}{D_{00}}}\mathcal{K}_1\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_p}\right)\left(1+\sqrt{G_{G1}}\left(\mathcal{K}_1-\mathcal{K}_2\right)\boxtimes\right)$	$H_{00}=\text{henry}\left(1-\frac{C_d}{6}\left(\mathcal{K}_1+\mathcal{K}_3+\mathcal{K}_4\right)\boxtimes\right)$	$\frac{e}{h}=\mathcal{K}_1\left(\frac{t_p\,q_p}{l_p^2\,m_e}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathcal{K}_1^2+\mathcal{K}_2^2\right)\boxtimes\right)$	$\frac{\mu_{de}}{8K}=\frac{2\pi}{8K}\left(1+\frac{16}{\rho_{up}}\left(\mathcal{K}_1^2+\mathcal{K}_3^2+\mathcal{K}_4^2\right)\boxtimes\right)$
$\Delta_{vcc}=\frac{E_1}{2\pi}\left(\frac{t_p}{l_p}\frac{m_p}{m_p}\right)\left(1-\sqrt{\zeta(2)}\,\mathcal{K}_\theta^{-2}\boxtimes\right)$	$m_-=\left(\frac{A}{16}\right)\frac{\text{GeV}}{\mathcal{K}_1}\left(\frac{t_p^2\,m_e}{l_p^2}\right)\left(1-3\left(\frac{A}{16}\right)\mathcal{K}_r^{-3}\boxtimes\right)$	$Z_0=4\pi\left(\frac{L_p^2\,m_p}{t_p^2}\right)\left(1+\frac{1}{2}Re\left(i^{l^{''}}\right)\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$m_3=\left(m_n-m_+\right)\left(1+\sqrt{\frac{4}{D_{00}}}\mathcal{K}_1\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$r_e=\mathcal{K}_1^2\left(\frac{L_p\,m_p}{t_p\,q_p}\right)\left(1-\sqrt{6\,G_{R1}}\left(\mathcal{K}_1-\mathcal{K}_2\right)\boxtimes\right)$	$\Omega_{00}=\text{ohm}\left(1-\frac{C_d}{6}\left(\mathcal{K}_1+\mathcal{K}_3+\mathcal{K}_4\right)\boxtimes\right)$	$\Phi_0=\frac{\pi}{\mathcal{K}_1}\left(\frac{t_p^2\,m_p}{l_p^2\,m_p}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathcal{K}_1^2+\mathcal{K}_2^2\right)\boxtimes\right)$	$\frac{\mu_{de}}{\mu_B}=\frac{2\pi}{8K}\left(\frac{m_e}{m_e}\right)\left(1+\frac{16}{\rho_{up}}\left(\mathcal{K}_1^2+\mathcal{K}_3^2+\mathcal{K}_4^2\right)\boxtimes\right)$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_p}{l_p^2}\frac{m_p}{m_p}\right)\left(1-\sqrt{\zeta(2)}\,\mathcal{K}_\theta^{-2}\boxtimes\right)$	$m_{be}=\frac{\sqrt{3}}{\mathcal{K}_1^2\,\mathcal{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_p^2}{l_p^2}\right)\left(1-\frac{3^{-1}}{\mathcal{K}_1\,\mathcal{K}_2}\left(\frac{s}{18}\right)\mathcal{K}_r^{-3}\boxtimes\right)$	$Z_p=\left(\frac{L_p^2\,m_p}{t_p^2}\right)\left(1+\frac{1}{2}Re\left(i^{l^{''}}\right)\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\mathcal{K}_1\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$a_0=\frac{1}{\mathcal{K}_1^2}\left(\frac{L_p\,m_p}{t_p}\frac{m_p}{m_e}\right)\left(1-\sqrt{6\,G_{R1}}\left(\mathcal{K}_1-\mathcal{K}_2\right)\boxtimes\right)$	$ST_0=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{K\,A_{mass}}{l_p^2\,T_p\,m_p}\right)^{3/2}\frac{K\,E_p}{T_p\,\rho_0}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathcal{K}_1\mathcal{K}_2}\right)\left(\mathcal{K}_1-\mathcal{K}_3-\mathcal{K}_4\right)\boxtimes\right)$	$\frac{h}{e}=\frac{\text{coulomb}}{\text{coulomb}}\left(\frac{L_p^2\,m_p}{l_p^2\,m_p}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathcal{K}_1^2+\mathcal{K}_2^2\right)\boxtimes\right)$	$b^{\prime}=\frac{v_{peak}}{2\pi}\left(\frac{1}{l_p\,T_p}\right)\left(1+\frac{16}{2\pi}\left(\frac{s}{s}\right)^2\left(\mathcal{K}_1^2-\mathcal{K}_3^2-\mathcal{K}_4^2\right)\boxtimes\right)$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_p}{l_p^2}\frac{m_p}{m_p}\right)\left(1-\sqrt{\zeta(2)}\,\mathcal{K}_\theta^{-2}\boxtimes\right)$	$G=\text{GeV}^2\left(\frac{t_p^2}{l_p^2}\frac{m_p}{m_p}\right)^2\left(1-\left(\frac{4\pi}{18}\right)\mathcal{K}_r^{-3}\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_p}{l_p}\right)\left(1+\frac{1}{2}Im\left(i^{l^{''}}\right)\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}^4}}\mathcal{K}_1\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$\sigma_e=\sqrt{\frac{32}{18}}\,2\pi\,\mathcal{K}_1^4\left(\frac{L_p\,m_p}{t_p}\frac{m_p}{m_e}\right)^2\left(1-2\sqrt{6\,G_{R1}}\left(\mathcal{K}_1-\mathcal{K}_2\right)\boxtimes\right)$	$ST_1=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{K\,A_{mass}}{l_p^2\,T_p\,m_p}\right)^{3/2}\frac{K\,E_p}{T_p\,\rho_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathcal{K}_1\mathcal{K}_2}\right)\left(\mathcal{K}_1-\mathcal{K}_3-\mathcal{K}_4\right)\boxtimes\right)$	$c_1=\left(2\pi\right)^2\left(\frac{L_p^4\,m_p}{t_p^3}\right)\left(1-\frac{4\pi^2}{32}\left(\mathcal{K}_1^2+\mathcal{K}_2^2\right)\boxtimes\right)$	$\kappa_B^{\prime}=\frac{1}{2\pi}\left(\frac{1}{l_p\,T_p}\right)\left(1+\frac{16}{2\pi}\left(\frac{2\pi}{s}\right)^2\left(\mathcal{K}_1^2-\mathcal{K}_3^2-\mathcal{K}_4^2\right)\boxtimes\right)$
$\text{Hz}:\text{eV}=\frac{2\pi\text{joule}}{\text{second eV}}\left(\frac{L_p^2\,m_p}{t_p^2}\right)\left(1+\sqrt{\zeta(2)}\,\mathcal{K}_\theta^{-2}\boxtimes\right)$	$G=\left(\frac{L_p^3}{t_p^3}\frac{m_p}{m_p}\right)\left(1-\left(\frac{4\pi}{18}\right)\mathcal{K}_r^{-3}\boxtimes\right)$	$A_{vel}=\mathcal{K}_1^2\left(\frac{L_p}{t_p}\right)\left(1-\frac{1}{2}Im\left(i^{l^{''}}\right)\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$\frac{\mu_e}{\mu_N}=\frac{2\,\mathcal{K}_1\,\mathcal{K}_2^2}{S}\left(\frac{m_e}{m_e}\right)\left(1-18\left(6\pi\right)\mathcal{K}_1\mathcal{K}_3\mathcal{K}_4\boxtimes\right)$	$\text{eV}:\text{A}_{mass}=\frac{\text{joule}^2}{\text{coulomb}}\sqrt{\frac{32}{18}}\frac{\mathcal{K}_1^6}{18\,\rho_{up}}\left(\frac{t_p^2\,q_p}{l_p^2\,m_e}\right)\left(1-\alpha_p\left(\mathcal{K}_1-\mathcal{K}_2\right)\boxtimes\right)$	$\sigma=\frac{3}{5}\frac{4\pi}{\Gamma(5)}\left(\frac{m_p}{t_p^3\,T_p^4}\right)\left(1+\sqrt{\frac{15}{8}}\left(\frac{2\pi}{\mathcal{K}_1\mathcal{K}_2}\right)\left(\mathcal{K}_1-\mathcal{K}_3-\mathcal{K}_4\right)\boxtimes\right)$	$c_{1L}=4\pi\left(\frac{L_p^4\,m_p}{t_p^3}\right)\left(1-\frac$	

$t_p = 5.39125836832313 \dots \times 10^{-48}$ s	Planck time
$l_p = 1.61625918175645 \dots \times 10^{-35}$ m	Planck length
$q_p = 1.87554596713962 \dots \times 10^{12}$ C	Planck charge
$T_p = 1.41678698590795 \dots \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \dots \times 10^{-8}$ kg	Planck mass
$\xi = 0.999999199376326 \dots \times 10^{-7}$	hyperbolic inversion boundary
5	number of dimensions
7	break in scale symmetry
$\kappa_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\kappa_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\kappa_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\kappa_4 = 1.87649603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\kappa_5 = -4.48762244916751 \dots$	hyperbolic radius constant
$\kappa_6 = 1.014462310924 \dots$	hyperbolic radian constant
$G_{61} = 1.01494160640965 \dots$	Gieseking's constant
$V = 0.02959831281930 \dots$	figure eight knot hyperbolic volume
$K = 2.09855944772190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domino tiling constant
$D_4 = 1.79162281206959 \dots$	dimer constant
$\alpha_1 = 0.764977018528596 \dots + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.259954037057192 \dots$	omega_2 constant
$i^{i^{i^{\dots}}}$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\phi_1 = (-1)^{1/3}$	imaginary golden ratio
$G_9 = 1.15872847301812 \dots$	tether length for grazing half unit circle
$\gamma = 3.14159265358979 \dots$	Archimedes' constant
$\gamma_{\text{sup}} = 2.29559714939263 \dots$	universal parabolic constant
$S = 5.24411510858423 \dots$	arc length of the unit lemniscate
$L_1 = 2.620557554292119 \dots$	lemniscate constant
$L_2 = 1.31102877174605 \dots$	1 st lemniscate constant
$L_3 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C = 0.847213084793979 \dots$	ubiquitous constant
$G_{60} = 0.836262841674073 \dots$	Gauss's constant
$C_{CF} = 0.697774567964819 \dots$	continued fraction constant
$C_{R1} = 1.928136044598509 \dots$	Ramanujan's 1 st continued fraction constant
$R_{\text{Fann}} = 0.2267107102035 \dots$	Fibonacci factorial constant
$M = -1.7175645643318 \dots$	Modeling constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$A = 1.2824712910062 \dots$	Glaser-Kinkelin constant
$S = 2.828252549678847 \dots$	Sierpiński's constant
$J_{0.1} = 2.040285557695772 \dots$	1 st root of the Bessel function
$C_{\text{RFP}} = 1.199678640255773 \dots$	Real fixed point of the hyperbolic cotangent
$D_{30} = 0.7390851332011560 \dots$	Dottie number
$L_{11} = 0.662743419349181 \dots$	Laplace limit
$\alpha_9 = 2.5920787509589 \dots$	alpha Feigenbaum constant
$\delta_9 = 4.66920160910299 \dots$	delta Feigenbaum constant
$P = 1.32471795724474 \dots$	plastic constant
$S = 3.24697960371714 \dots$	silver constant
$C_0 = 0.624543657196732 \dots$	QRS constant
$C_{\text{QSA}} = 2.03816937970215 \dots$	associated QRS constant
$\chi_{\text{HSM}} = 0.353236371854995 \dots$	Haflner-Sarnak-McCurley constant
$T_0 = 2.73150000000000 \dots \times 10^5$ kelvin	freezing point of water
$p_0 = 1.00000000000000 \dots \times 10^2$ pascal	1 bar
$p_1 = 1.013250754773 \dots \times 10^{-15}$ m	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{24}$ joules	energy associated with ΔE_{cs}
$f_1 = 5.40000000000000 \dots \times 10^{14}$ Hz	frequency of max/light wavelength
$\nu_{\text{peak}} = 4.96511423714427 \dots$	peak k of spectral radiance/unit conversion
$\nu_{\text{peak}} = 2.82149397212207 \dots$	peak emission of spectral flux/unit frequency
$\log(x) =$ natural logarithm	$\text{lm} =$ lumen = cd · sr
$\Gamma(x) =$ gamma function	$N =$ newton = m kg/s ²
$\zeta(x) =$ Riemann zeta function	$\mathcal{N} =$ noether = m kg/s
$\eta =$ derangement function	$\text{ohm} = \text{m}^2 \text{ kg/s}^2 \text{ C}^2$
units	composite units
m = meter	A = ampere = C/s
s = second	F = farad = s ² C ² /m ² kg
K = kelvin	Hz = hertz = 1/s
kg = kilogram	J = joule = m ² kg/s ²
derived units	
MHz = megahertz = 1.00000000000000 $\times 10^6$ 1/s	
f = femtometer = 1.00000000000000 $\times 10^{-15}$ m	
eV = electron-volt = 1.00000000000000 eV	
MeV = megaelectron-volt = 1.00000000000000 $\times 10^6$ eV	
GeV = gigaelectron-volt = 1.00000000000000 $\times 10^9$ eV	
$\tau_{\text{e}} = \left(\frac{4\pi}{95}\right)^2$	$\tau_{\text{n}} = R$
$\tau_{\text{e}} = \left(\frac{4\pi}{95}\right)^2$	$\tau_{\text{n}} = \sqrt{\frac{C_0}{U}}$
	$\tau_{\text{de}} = \sqrt{\frac{C_0}{6} \kappa_1 \kappa_2}$
	$E_p = \frac{L_p^2 m_p}{\ell_p^2}$
external transform space	Physics Honorary Thad Roberts 
$v_{24} = \left(\frac{4\pi}{51}\right) \left(\frac{4\pi}{15}\right) \left(\frac{4\pi}{35}\right) \left(\frac{4\pi}{18}\right)^2 \left(\frac{4\pi}{32}\right) \left(\frac{4\pi}{8}\right)^4$	September-22-2025
figure eight knot hyperbolic volume	Angela Arvizu Vanessa Moon Matthew Fox David Heggeli Eschelein Azha Av